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Wind protection rather than soil water availability contributes to the restriction of high-mountain forest to ravines

J. Sparacino[1] · D. Renison[2] · A. M. Devegili[3] · R. Suarez[4]

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Abstract

Tropical and subtropical high-mountain forests are often restricted to ravines, with forest presence being rare in other topographies such as ridges. We tested the hypothesis that this distribution is partly due to increased soil water content and/or protection from winds in ravines as compared to ridges. We planted 396 saplings of the dominant tree species in a three-factor experiment that included topography (ridge/ravine), wind protection (protected/unprotected) and water addition (watered/not watered). We monitored wind speed and soil water content for one complete annual cycle and sapling survival and growth for 27 months. Sapling survival presented marginally significant differences for topography, being slightly greater in ridges than in ravines (0.88 and 0.78, respectively), with no significant differences for wind protection or water addition treatments. By contrast, sapling growth was considerably reduced in ridges as compared to ravines (16.1 and 26.0 cm, respectively); the wind protection treatment showed a significant positive effect on growth (saplings grew 25.5 and 16.5 cm with and without wind protection, respectively), whereas the water addition treatment had no effect. Importantly, saplings under wind protection treatment in ridges had similar growth to that of unprotected saplings in ravines, and growth difference due to the wind protection treatment was positively correlated with wind intensity, suggesting that reduced wind impact in ravines is an important abiotic factor that promotes forest occurrence in ravines.

Keywords Woodland · Wind shelter · Watering · *Polylepis australis* · Planting experiment

✉ J. Sparacino
javiersparacino@gmail.com

1 Instituto de Física Enrique Gaviola, Facultad de Matemática, Astronomía, Física y Computación (CONICET - Universidad Nacional de Córdoba), Av. Medina Allende s/n, X5000HUA Córdoba, Argentina

2 Instituto de Investigaciones Biológicas y Tecnológicas, Centro de Ecología y Recursos Naturales Renovables (CONICET - Universidad Nacional de Córdoba), Av. Velez Sarsfield 1611, X5016GCA Córdoba, Argentina

3 Laboratorio Ecotono, INIBIOMA (CONICET - UNCOMA), Pasaje Gutiérrez 1125, 8400 Bariloche, Argentina

4 Proyecto de Conservación y Reforestación de las Sierras de Córdoba, Ortiz de Ocampo 741, Río Ceballos, Córdoba, Argentina

Introduction

In subtropical and tropical mountain ranges, forests and woodlands (hereafter "forests") above the timberline often occur in ravines, with other topographies such as ridges, convex summits and flat highlands having little or no tree cover (Körner 2012). The causes of this peculiar restriction of forests to ravines have been a matter of debate, and it has become increasingly clear that multiple factors explain this distribution (Renison et al. 2018). In the subtropics and tropics, the area above the continuous timberline is within the "ecosystems uncertain" climate envelope, where it is not possible to accurately predict the occurrence of grasslands and forests based only on climatic conditions (Whittaker 1975). These areas may be regarded as "consumer-controlled" areas, in which large herbivores and fire, the latter considered an abiotic consumer, also play an important role in determining the presence of grasslands or woodlands (Bond 2005).

Livestock and fires are two major determinants of the prevalence of forests in ravines in many subtropical and tropical mountain ranges above the timberline (Kessler 2002). Fire incidence is higher outside ravines due to the tendency of fire to move more easily uphill than downhill (Renison et al. 2006). Fires scorch the aerial part of trees, increasing mortality and basal sprouting. Livestock are then attracted to the post-fire regrowth and browse basal sprouts, further reducing their survival and growth. This keeps high-altitude burnt forests at an arrested or early successional stage (Alinari et al. 2015; Argibay and Renison 2018; Giorgis et al. 2010). Since fire ignitions have been mainly attributed to humans and livestock densities are kept high due to management, the prevalence of forests situated within ravines is currently thought to be caused by humans (Kessler 2002; Renison et al. 2018).

Abiotic site factors are also likely to influence forest distribution in subtropical and tropical mountain ranges above the timberline, especially when the overriding influence of domestic livestock and human-ignited fires is reduced. In particular for the tropical Andean mountain range, it has been postulated that forests above the continuous timberline are situated in ravines because these sites provide high soil water content and protection from strong winds and low-freezing temperature (Kessler 2002; Rada et al. 2009). Accordingly, in subtropical central Argentina establishment success in a seeding and planting experiment was higher in ravines than in upper topographies, even in the absence of human-caused disturbances (Renison et al. 2015), although the abiotic factors explaining the pattern were not investigated. Understanding the abiotic site factors that reduce tree species establishment outside ravines could help determine the extent to which forests are able to colonize other sites if fire and livestock are excluded or reduced.

Water availability is an important limitation to plant growth in dry and seasonally dry regions such as a large portion of the subtropical and tropical Andean range. Water in soil tends to drain and accumulate in lower topographical positions, where it could be more available for plant growth (Cingolani et al. 2015). Slope is also an important component of soil water availability, with gentler slopes accumulating more water than steeper slopes, given similar upslope area drainage (Sørensen et al. 2006). Additionally, the exposure to winds in open sites, including upper topographies, as compared to protected ravines, might cause damage to leaf parts and increase evapotranspiration (Gardiner et al. 2016). In particular, separating the contribution of soil water content from exposure to wind as causes for the low growth rates of trees in upper topographies is crucial for understanding the potential of forests to colonize sites other than ravines when fires and livestock are excluded. If soil water content is more limiting than wind, it would be expected that upper topographies will be able to sustain only a limited number of trees. Instead, if exposure to winds is more limiting than water availability, this would have important implications because forests, when present, might reduce wind speed and create a microclimate where saplings and juveniles could thrive, thus making further regeneration possible. Other possible influences on sapling and juvenile performance at the microsite scale are varying soil depths, light availability and the presence of accompanying vegetation that may compete with or facilitate tree establishment, depending on context and life-stage (e.g. Loranger et al. 2017; Schönbeck et al. 2015).

In subtropical and tropical South America, species of the genus *Polylepis* (Rosaceae) dominate the canopy of the forests above the timberline along the Andes and associated mountains. Thus, the discussion about the causes of forest restriction to ravines has been termed “the *Polylepis* problem” (Kessler 2002). The *Polylepis* forest belt extends from Venezuela to central Argentina along 5400 km, and consists of a mosaic of small forest patches embedded in a grassland matrix. These high-altitude forest patches are usually dominated by one of the approximately 28 *Polylepis* species, whose taxonomy and ecology are poorly known (Renison et al. 2018; Segovia-Salcedo et al. 2018). High-altitude tropical and subtropical mountain ranges are rare outside South America and only include small areas of Africa, New Guinea, and the southern ranges of the Himalayas. To our knowledge, there are no studies describing high-altitude forest distribution in relation to topography in those areas out of South America.

In this work, we aimed at examining the influence of soil water limitation and wind exposure to survival and growth of tree saplings in upper topographies using *Polylepis australis* (Bitter) in the mountains of central Argentina as a case study. We tested the hypothesis that the restriction of forests to ravines could be partly explained by increased soil water content and/or protection from winds within ravines as compared to upper topographies such as ridges. We predicted that survival and growth of saplings in the upper topographies would be lower than in ravines, but this difference would be compensated by experimentally increasing soil water content, reducing wind speed or both. If the prediction were true, the presence of forests might be possible in both ravines and upper topographies, although forest recovery after disturbances such as livestock browsing and fire would presumably take longer in upper topographies. Moreover, if survival and growth of saplings were reduced by water availability, wind, or both factors, reforestation efforts in upper topographies could be enhanced by watering or protecting saplings from the wind. Additionally, we also aimed at identifying which microsite conditions (sun exposure, inclination, soil depth, vegetation cover, and vegetation height) could be affecting sapling performance.

Materials and methods

Study area and species

The study was conducted in the upper portion of the mountains of central Argentina, locally named “Sierras Grandes de Córdoba” (north–south orientation; up to 2884 m a.s.l.; 31°34′S; 64°50′W). At 2400 m a.s.l., mean temperature is 6.7 °C in the dry and cold season (from May to September) and 11.2 °C in the wet and warm season (from October to April), with below-zero temperatures being possible year-round (Colladon and Pazos 2016). Mean annual precipitation from 1992 to 2012 averaged 1008 mm, with 90% of the rainfall occurring in the warmer months (Colladon and Pazos 2014). Livestock rearing is the main economic activity in the area. The landscape consists of a mosaic of forest patches (at present, about 12% of the area), tussock grasslands, short grazing lawns, rocky outcrops, and exposed rock surfaces. Forest patches are dominated by *Polylepis australis,* which at lower altitudes mixes with other woody species, mainly *Maytenus boaria* and *Escallonia cordobensis* (Cabido et al. 2018; Cingolani et al. 2004, 2008). Fires affect approximately 1.3% of the area (7421 ha) each year, with 73% of them covering less than 100 ha (Argañaraz et al. 2015).

Polylepis australis is endemic to the mountains of central and north-western Argentina; individuals can reach 14 m in height and livestock at stocking densities typical of central Argentina browse on average 98% of the accessible shoots (Giorgis et al. 2010; Pollice et al. 2013; Renison et al. 2011). *P. australis* saplings under livestock exclusion grow approximately 8 cm per year (Simoes Macayo and Renison 2015); therefore, it takes more than 18 years for them to grow tall enough to escape browsing of their tallest growth buds.

Experimental design

A planting experiment was performed in the northern part of the mountains of central Argentina (Córdoba province, Pampa de Achala Water Reserve; 2280 m a.s.l.; 31°26.4′S; 64°48.52′W; see Fig. 1a, b). The experimental design consisted of a three-factor experiment that included topography (ridge/ravine), wind protection (protected/unprotected) and water addition (watered/not watered). A total of 396 saplings were manually planted in six sites (64 in each of three ridge sites and 68 in each of three ravine sites). Numbered saplings were distributed irregularly within each site at a minimum distance of 5 m from each other. The wind protection and water addition treatments were assigned in a completely randomized design. The rugged topography of the area allowed us to choose ridges and ravines that were within short distances; thus, other possible confounding factors such as altitude, latitude or biotic factors like possible local insect outbreaks were controlled.

The six sites were located in a 45-ha area that was fenced to exclude livestock in 2013 (Fig. 1c). The experimental plantings were performed in January and February 2015, and all saplings were watered at planting (6 l), independently of the water addition treatment. Outplanting was performed during the rainy season in order to optimize survival (Renison

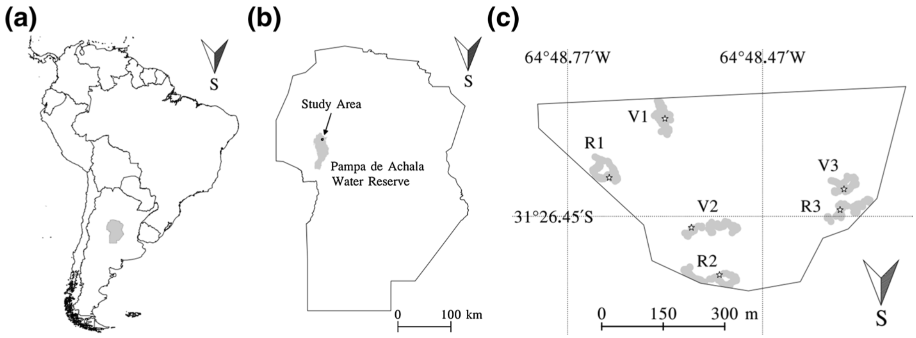


Fig. 1 Location of the study area: **a** Córdoba province in Argentina and South America; **b** Pampa de Achala provincial Water Reserve in Córdoba province; **c** study sites: the fencing of the livestock exclusion area is represented by a continuous line and the six study sites where saplings were planted are indicated in gray. Ridges are labeled with an 'R' and ravines with a 'V', they are numbered in pairs. The locations of the six Hobos that recorded hourly temperatures are represented by stars

et al. 2002). On the outplanting date, saplings were 9 months old and 14.2 ± 0.4 cm tall (mean height ± standard error).

Wind protection treatment

The wind protection treatment consisted of discarded transparent 6.25-l water bottles cut at both ends to have an almost cylindrical protection of 22 cm in height and 16 cm in diameter. The protection was held in place by tying it to a 60-cm long pole that was buried in the same hole as the sapling and to the 60-cm long numbered metal pin that identified saplings (Fig. 2). A 36-cm^2 square outlet was cut at the lower side of every plastic protection to allow the escape of small animals that might fall in. The wind protections provided partial reduction of wind impact, as they were on average 8 cm taller than the saplings at planting and were 17 cm lower than saplings on the last monitoring date. We used discarded water bottles because we considered that their larger diameter would reduce or avoid heat accumulation as compared to commercial wind protectors; we also considered them more likely to be used by forest restoration practitioners than commercial options, given that funding available for forest restoration in *Polylepis* forests distribution area is usually scarce. All wind protections remained in place during the monitoring period and none of them were lost; they were all removed at the end of the experiment.

Water addition treatment

The water addition treatment consisted of watering through a 60-cm long tube which was partially buried next to the sapling, to a depth of 35 cm (Fig. 2). Thus, water could be applied directly to the sapling roots, reducing water loss as compared to conventional surface watering. Treated saplings were watered with 700 cm^3 of water on 20 occasions throughout 1 year, from May 7, 2015 to May 6, 2016, approximately every 14 days during the warm months (October to April) and every 28 days during the cold months (May to September). The amount of water was chosen in order to provide each sapling with an extra 50% of the total amount of rainwater, according to the annual average for the region (1008 mm, Colladon and Pazos 2014), assuming a water collection diameter of 20 cm and 22 watering events. Two watering events were not performed due to intense precipitations on the scheduled dates; therefore, the actual amount of added water was 44% of the total amount of rainwater (446 mm).

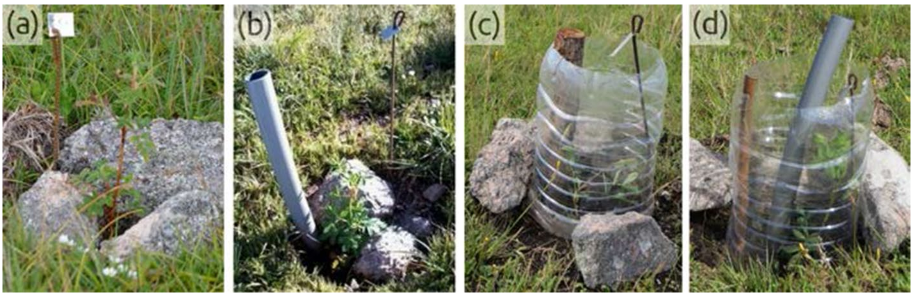


Fig. 2 Saplings planted under different treatments: **a** control, **b** water addition treatment, **c** wind protection treatment, **d** water addition and wind protection treatments

Site characterization

Each one of the 396 microsites (30-cm radius area around each sapling planting site) was characterized in terms of altitude, sun exposure, general slope inclination, soil depth, vegetation cover and vegetation height. Altitude above sea level was determined using GPS (eTrex 20, Garmin). Sun exposure was estimated using a hand clinometer as the portion of the sun trajectory that is not covered by topography or vegetation and measured in degrees (in a perfectly flat area, sun exposure would be 180°, usually in mountain regions it is less than 180°, this is particularly true in ravines). Soil depth was estimated by hammering a 1-m long pin into the soil until it was judged that the underlying rock was reached. Note that soil depth was underestimated whenever it was greater than the pin length. Vegetation cover was estimated visually as the percentage of the microsite covered by plants. Vegetation height was estimated as the average of three random heights as measured with a tape measure from the ground to the highest stem or leaf. We measured variables for microsite characterization between January and February 2015.

Mean annual temperatures and absolute minimum/maximum temperatures were obtained from six Hobo data loggers, with one placed at each site (positions are depicted with stars in Fig. 1c). Temperature was measured every hour for 1 year from February 23, 2015 to February 22, 2016.

Precipitation was 502 mm during the January–February 2015 period, almost 60% higher than the 1992–2012 average for the same period. Between May 7, 2015 and May 6, 2016, when the water addition treatment was performed, precipitation was 1012 mm, which is very close to the average of 1008 mm for the 1992–2012 period (Colladon 2018).

Measurements

Soil water content and wind intensity

Soil water content was measured at each planting microsite using a Moisture Probe Meter (MPM160, ICT International Pty Ltd.). Wind intensity measurements (between 6 and 12 per site) were performed using an Anemometer MultiMeasure Basic (BA05, TROTEC GmbH & Co. KG). Each measurement consisted of assessing wind intensity while walking across the area of approximately 10 sapling microsites and recording wind direction and maximum intensity. Soil water content and wind intensity were measured every 14 days for 1 year, from February 7, 2015 to February 6, 2016.

Sapling performance

We recorded sapling height at planting in January and February 2015, and sapling survival and height in April 2015, August 2015, November 2015, February 2016, May 2016, and March 2017 for a total of 27 months after planting and corresponding to three growing seasons. Saplings were considered dead when stems were dry and no green leaf or sprout was observed. Survival was updated over time for those saplings that were apparently dead but presented new sprouts in the next surveys. Height was measured from the ground to the terminal bud. Growth was calculated as the difference between height recorded on the dates of measurement and height at planting.

Browsing

Although the experimental sites were inside a fenced area intended for livestock exclusion, the presence of cattle was noticed during the experiment. When this occurred, the date and number of cattle were recorded, and cattle were chased out of the fences, which were later checked for possible damage and repaired when necessary.

In order to characterize livestock presence inside the exclusion area, an upper limit to the stocking density was calculated by summing up the number of cattle observed multiplied by the maximum time they could have been inside the area. This period was estimated assuming livestock had been inside the exclusion area the whole time since the last time we visited the area and found no livestock there. If n_v is the number of visits in which cattle were observed inside the exclusion area, N_i is the number of cattle inside the area on visit i and t_i is the time (in days) since the last visit, then an upper limit to the stocking density, ρ_u, can be calculated as follows:

$$\rho_u = \frac{1}{T \cdot A} \sum_{i=1}^{n_v} N_i \cdot t_i \tag{1}$$

where T is the 27-month monitoring period (in days), and A is the fenced area (in ha). Thus, the stocking density inside the exclusion area, ρ, measured in cattle equivalents ha^{-1}, will have a value lower than the calculated upper limit ($\rho < \rho_u$), which provides a useful reference value for comparison with other livestock loads in the region. When monitoring saplings for survival and height, we also recorded browsed saplings as the percentage of buds with cut stems, which is evidence of consumption by livestock.

Statistical analysis

We performed Generalized Linear Mixed Models (GLMMs) to analyze the response variables survival (assuming a binomial distribution) and growth (assuming a normal distribution and only considering saplings that survived to the last monitoring date). Site (1, 2, or 3) was considered a random factor, and topography (ridge or ravine), water addition (watered or not watered), and wind protection (protected or unprotected) were included as fixed factors. We also analyzed all possible double and triple interactions.

Because sapling survival and growth turned out to have opposing patterns of results, thus making their interpretation difficult, we integrated both measures into an indicator of sapling performance by multiplying mean growth for the relevant treatments by the respective survivals (as in Renison et al. 2015).

To better interpret the results, we also calculated growth difference due to the wind protection treatment as the differences between growth with wind protection minus growth without wind protection, and correlated the differences with the average wind speed of each site using Pearson correlation. A similar analysis was performed for the water addition treatment and Pearson correlation with soil water content was calculated.

To analyze the relationship between growth and herbivory, we used the maximum percentage of browsing recorded for each sapling as an indicator of herbivory pressure. Incidence of herbivory among factors (topography, water addition and wind protection) was analyzed using Fisher's exact test; for this test, the response variable browsing was transformed to a nominal binary variable (browsed at least once and unbrowsed). Growth was analyzed again for the subpopulation of unbrowsed saplings to avoid any possible spurious interpretation of the results.

In order to compare abiotic and biotic characteristics of the sites, GLMMs were performed to analyze sun exposure, slope inclination, soil depth, vegetation cover, and vegetation height. Each variable was analyzed assuming a normal distribution, with site (1, 2, or 3) as a random factor and topography (ridge or ravine) as a fixed factor. To better explore possible explanatory variables, another GLMM with growth as response variable was performed, as the one described at the beginning of this section but adding as covariates the relevant microsite measurements we estimated at each sapling site: sun exposure, slope inclination, soil depth, vegetation cover and vegetation height.

In all cases, results were considered significant at $P<0.05$. We checked residuals for normality and homoscedasticity. All statistical analyses were performed using R software environment (version 3.4.2, R Core Team 2018).

Results

Sapling performance, wind intensity and soil water content

Survival fraction of saplings presented marginally significant differences for topography, with survival being 13% greater in ridges than in ravines (0.885 ± 0.050 and 0.784 ± 0.081, respectively), and with no significant difference for the water addition or wind protection treatment, or their interactions (see Table 1 and Fig. 3).

Sapling growth was 38% lower in ridges than in ravines (16.14 ± 2.76 and 26.00 ± 5.58 cm, respectively). The wind protection treatment increased sapling growth by about 50%, from 16.51 ± 1.08 cm in the control saplings to 25.46 ± 1.60 cm in the wind protected saplings, whereas the water addition treatment and the interactions were non-significant (see Table 1 and Fig. 3).

Wind intensity was twice higher in ridges than in ravines (5.76 ± 1.09 and 2.79 ± 0.09 m s^{-1}, respectively, for annual averages). Notably, growth difference due to the wind protection treatment was positively correlated with average wind intensity, as measured for each of the six sites ($r=0.8319$, $t=2.9988$, $P=0.0400$, Fig. 4). Contrary to our expectations, soil water content was similar in ridges and ravines (31.54 ± 5.13 and 29.55 ± 0.75%, respectively) and growth difference due to water addition treatment

Table 1 Results of generalized linear mixed models for survival (assuming a binomial distribution) and growth (assuming a normal distribution) of *P. australis* saplings, testing the effect of fixed factors: topography, water addition treatment (WaT) and wind protection treatment (WiT), as well as all possible double and triple interactions

	Survival (fraction)		Growth (cm)	
	z	P	t	*P*
Topography	−1.9445	0.0518	3.4827	**0.0006**
WaT	−0.9096	0.3630	0.6924	0.4892
WiT	0.7446	0.4565	3.3724	**0.0008**
Topography × WaT	1.7027	0.0886	−0.2797	0.7799
Topography × WiT	0.2326	0.8161	−0.7164	0.4743
WaT × WiT	−0.2269	0.8205	0.0259	0.9793
Topography × WaT × WiT	−1.4739	0.1405	−0.4284	0.6686

Significant results ($P<0.05$) are shown in bold

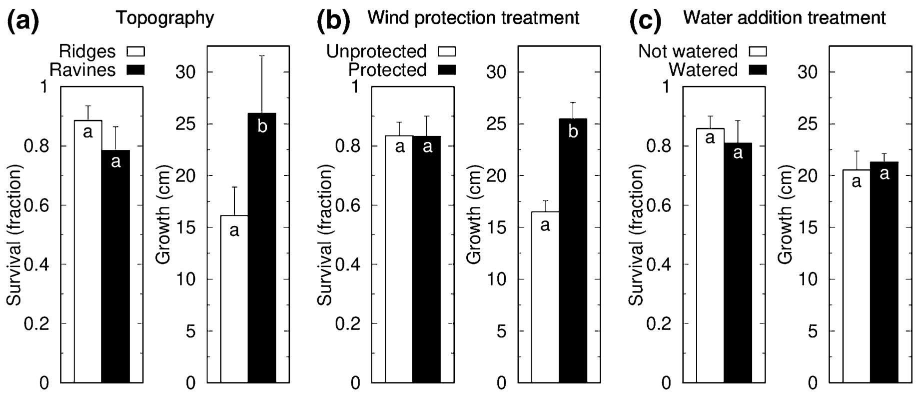
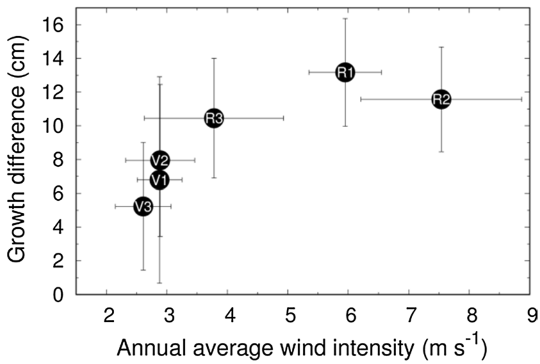


Fig. 3 *Polylepis australis* sapling survival and growth as a function of **a** topography, **b** wind protection treatment, **c** water addition treatment. Error bars indicate standard errors. Different letters indicate significant differences at $P<0.05$

Fig. 4 Growth difference between saplings with and without wind protection treatment as a function of the annual average wind intensity. Each dot corresponds to measurements of a different site, as indicated by the labels in the plot ('R' stands for ridge and 'V', for ravine). Error bars are the result of standard errors propagation



was not correlated to the average soil water content of each of the six sites (r = 0.0847, t = 0.1700, $P=0.8732$). Results of both wind intensity and soil water content for a complete annual cycle are shown in Figs. 5 and 6, respectively ("Appendix 1" section).

An integration of sapling survival and growth into a single performance indicator (survival × growth) showed that growth patterns were the main drivers of the performance indicator. The sapling performance indicator was 30% lower in ridges than in ravines (an average growth of 14.28 and 20.38 cm per planted sapling, respectively). Similarly, the wind protection treatment increased the sapling performance indicator by 54% (average growth of 21.21 cm per planted sapling with wind protection and 13.77 cm per planted sapling without wind protection). Instead, sapling performance indicator was 2% lower with water addition treatment (average growth of 17.23 and 17.65 cm per planted sapling with and without water addition, respectively).

Browsing

Estimated average livestock density within the exclosure was less than 0.011 cattle ha^{-1} according to the upper limit calculated following Eq. 1. Despite the low livestock density, browsing involved a total of 154 saplings. Survival of browsed and unbrowsed saplings was similar ($P=0.5892$, Fisher's exact test for browsed and unbrowsed saplings). Instead, sapling growth and browsing were negatively correlated (-0.3988, $t=7.8766$, $P<0.0001$). Browsing was greater in ridges than in ravines ($P=0.0003$) and in saplings without than with wind protection ($P<0.0001$), but was similar between saplings with or without water addition ($P=0.6579$).

Even though browsing was not equally distributed among factors, sapling growth results are robust: when we repeated the analysis of growth including only the unbrowsed saplings ($N=176$, see Table 3 in "Appendix 2" section for detailed distribution), results were similar to those obtained for the complete set of saplings. Height growth was greater in ravines than in ridges ($t=3.2362$, $P=0.0015$), showing significant differences for the wind protection treatment ($t=2.4200$, $P=0.0166$) and no significant differences for the water addition treatment ($t=-0.6643$, $P=0.5074$).

Site characteristics

The abiotic and biotic characteristics of the sites are shown in Table 2. As a consequence of their topographic position, sun exposure was 50% greater in ridges than in ravines. Ridges also had gentler slopes and almost 30% shallower soil depth than ravines. Ridge sites had 10% less vegetation cover than ravines, although vegetation cover was above 80% in both, and vegetation height was, on average, 60% lower in ridges than in ravines.

The model that included the three main factors (topography, wind protection and water addition), and the microsites measurements as covariates, showed significant differences in height growth for the wind protection treatment but not for the water addition treatment (same pattern as the model without microsite conditions as covariates). Differences in topography became non-significant as compared to the model without covariates, whereas sun exposure, a covariate closely related to topography, became significant with more height growth at microsites with less sun exposure. Vegetation height is the only other covariate that presented a significant positive association with sapling height growth. Vegetation cover, slope inclination and soil depth were not significant (see Table 4 in "Appendix 3" section for details of this statistical analysis).

Discussion

Our case study represents the first experimental support of the hypothesis that the restriction of high-mountain forests to ravines is partly explained by wind protection. Interestingly, water availability would not be a contributing factor even when our study area has a long dry season. More case studies in other tropical and subtropical high-mountain forest will be needed to determine the generality of our findings.

Plants are known to respond to increased wind loading by acclimation at different structural scales from the cell to the whole plant (Gardiner et al. 2016, and references therein). Tree responses to wind include reduced elongation growth and increased radial growth (Ennos 1997; Lawton 1982; McArthur et al. 2010). Thus, wind is a major factor influencing tree growth in other species, as shown here for *P. australis* saplings. Wind also affects plant evapotranspiration and temperature; therefore, it is not easy to differentiate mechanosensitive effects from water stress or cooling effects (Gardiner et al. 2016). However, as we

Table 2 Abiotic and biotic characteristics of the sites used for the planting experiment

	Ridges	Ravines	Statistical analyses	
			t	*P*
Mean annual temperature (°C)	10.6 ± 0.4	10.23 ± 0.4		
Absolute minimum temperature (°C)	− 9.5	− 7.8		
Absolute maximum temperature (°C)	35.3	41.5		
Altitude (m a.s.l.)	2209–2306	2196–2291		
Sun exposure (°)	154.9 ± 10.2	104.41 ± 11.9	− 32.1324	< 0.0001
Slope inclination (°)	8.77 ± 3.15	18.64 ± 2.79	11.4560	< 0.0001
Soil depth (cm)	48.4 ± 11.1	71.8 ± 2.2	8.6995	< 0.0001
Vegetation cover (%)	81.7 ± 7.3	90.5 ± 1.1	4.8957	< 0.0001
Vegetation height (cm)	6.34 ± 0.62	16.42 ± 5.80	12.1954	< 0.0001

Temperatures (mean and absolute minimum/maximum) were obtained from the Hobos placed in each site. Minimum and maximum values are reported for altitude, as obtained with a GPS at the lowest and highest sapling in each area. Averages and standard errors are reported for sun exposure, slope inclination, soil depth, vegetation cover, and vegetation height estimated for each planted sapling

found no significant interaction between wind protection and water addition treatments, our results suggest that enhanced evapotranspiration due to wind exposure would not be the physiological mechanism involved in limiting *P. australis* growth. It has been shown that, as long as water is available, even strong winds do not critically affect the water balance of trees (Holtmeier and Broll 2010) and an increase in wind can even decrease their transpiration rate (Grace 1988). For instance, a wind-exposure experiment with *Cecropia schreberiana* saplings produced several responses, including a reduction in plant height, but had no effects on transpiration rate or water-use efficiency (Cordero 1999).

The importance of wind exposure in reducing sapling growth was further supported by the positive correlation found between the effect of wind protection and wind speeds at the site level. Similar results were found in other situations; for example, annual height growth of *Pinus cembra* was highly and negatively correlated with wind speed in a 25-year experiment in Austria on an afforestation area where water is generally not a limiting factor. Interestingly, annual height growth at the most wind-exposed site improved with time due to the closure of the stand (Kronfuss and Havranek 1999). Thus, restoration in upper topographies by means of tree planting might help to create better microclimate conditions for the recruitment of new seedlings, and an increase in annual growth might be expected in the long term as the forest stand begins to close. Similar growth rates of saplings with the wind protection treatment in ridges to those of unprotected saplings in ravines suggest that the restoration of upper topographies could be feasible and equally fast as in lower topographies if trees are protected from wind. It should be noted that the wind protections we used were 17 cm shorter than average sapling height at the end of the experiment. Presumably, taller protections would have made these saplings grow even more than they did.

Wind protections were made of discarded water bottles; therefore, another possible effect other than wind reduction could be the formation of a greenhouse-like microclimate within the plastic shelter favorable for sapling growth and survival (Hammatt 1998; Kjelgren and Rupp 1997; Oliet and Jacobs 2007; Potter 1991). We did not measure temperature inside the protections. We judged that air could circulate freely across the shelters because they are wide enough (protections are 22 cm tall and 16 cm wide) and all had an outlet cut at the lower side section, thus preventing heat accumulation and microclimate formation inside them (Bergez and Dupraz 2000).

Water can be limiting for plant growth in environments under harsh conditions, like mountain ranges (Littell et al. 2008). However, in our study we did not find any response to water addition and, unexpectedly, soil water content was similar in ravines and ridges, supporting the idea that *P. australis* restriction to ravines would be explained by others factors. The lack of *P. australi*s response to water addition could be due to the better competitive ability of surrounding grasses and forbs to capture water (Casillo et al. 2012), even considering that we watered at the sapling root level. The similarity in soil water content in ravines and ridges was probably due to the much steeper slopes of the study ravines than those of ridges, a scenario that also occurs in most ravines occupied by *Polylepis* (Cingolani et al. 2004). Gentler slopes are known to accumulate more water (Sørensen et al. 2006), which could be compensating the lack of upslope water flow in ridges, as does happen for lower topographical positions (Cingolani et al. 2015). Rainfall was very close to the historical average for the region between May 2015 and May 2016, when the water addition treatment was performed, but during the previous period (January and February 2015) precipitation was almost half of the annual average (60% more than the average for the 2 months together). This precipitation in the first growing season could have been high enough to reduce potential differences in storage capacity between ridges and ravines. Perhaps under drier conditions, differences between watered and not watered saplings could become significant. In the highlands of northern Chile, watering was reported to be important for *Polylepis tarapacana* plantations in an area with average annual rainfall of 123 mm (Orrego 2011).

Several microsite conditions proved to be significantly different between ridges and ravines, and could be playing an important role in *P. australis* performance in upper topographies. Vegetation height showed a positive association with sapling growth. Many facilitative interactions have been reported between plants (Brooker et al. 2008; Padilla and Pugnaire 2006) and a possible interpretation of the positive association between vegetation height and sapling growth could be that surrounding vegetation acted as a wind shelter for the experimental saplings. However, more research is needed to determine the role of accompanying vegetation in the wind protection of saplings, especially because this aspect of our study was only descriptive but not experimental. While soil depth was significantly greater in ravines than in ridges, it presented no significant effect on sapling height growth (data not shown). Although it could play an important role in the long term, when saplings grow into adult trees, soil depth does not seem to be important at the early stages of saplings. The negative association between sun exposure and sapling growth is most likely due to the high association between topographic position (ridge or ravine) and sun exposure, where ravines receive much less sun light than ridges, and may be capturing other conditions, such as wind intensity. Our study species is considered an early successional species tolerant to high light conditions and in several previous studies it has been well documented that exposure to sun per se does not influence sapling or adult tree success (Renison et al. 2005, 2011; Suarez et al. 2008).

Although livestock was detected inside the study area, livestock density was much lower than the stocking densities typical of the mountains of central Argentina (Von Müller et al. 2012). The presence of livestock inside the area was in part due to the rugged topography, which facilitated livestock entry at certain points, but also due to intentionally broken fences, which allowed livestock access to better pastures within the exclosure, especially during the dry season. When cattle were detected inside our study area, the fence was carefully checked and repaired or reinforced. Since rugged topography and livestock rearing in the neighboring areas are typical of many mountains, the occasional presence of livestock in study exclosures could be more common than normally acknowledged. The greater incidence of browsing in ridges than in ravines in central Argentina was reported by Alinari et al. (2015) and could be attributed to large-herbivore avoidance of ravines, where they are more vulnerable to predators (Pia et al. 2013; Ripple and Beschta 2003). Herbivory also had greater incidence on saplings without wind protection, probably because plastic shelters act as an obstacle preventing livestock access to buds. Even though livestock differentially browsed some of our study saplings, the results are robust enough, as shown when the analysis was repeated for the subgroup of saplings that were never browsed. It is also worth noting that the reduced growth rates caused by wind extend the period of vulnerability of saplings to terrestrial herbivores and fires, thus increasing the difficulties of forest regeneration and restoration attempts in upper topographies.

Overall *P. australis* sapling mortality was not spatially clustered and survival fraction was high (average fraction of 0.8333) and very similar among factors, as reported for other planting experiments involving this species (Renison et al. 2005, 2015). Thus, growth patterns are the main drivers of sapling performance and growth improvement is important to optimize restoration efforts.

Conclusions and implications for practice

Of the two abiotic factors that we tested in this experiment, wind protection but not soil water availability contributed to the restriction of high-mountain forest to ravines. Those abiotic factors were chosen under the hypothesis that wind is more intense in ridges and that water tends to drain to lower topographies, like ravines, where it is more available for trees to grow. On the one hand, soil water content in ridges was found to be comparable to that of ravines. This finding, along with the gentler slopes of ridges which might enhance water retention, helps us to understand that water addition treatment yielded non-significant results. On the other hand, winds are more intense in ridges than in ravines, and the proposed wind protection treatment was effective in increasing sapling height growth, more so at sites with stronger wind. We conclude that wind protection within ravines is one of the abiotic factors contributing to forest restriction to ravines in central Argentina and possibly in other subtropical and tropical high-mountain forests.

Our results lead to three main implications. First, our study supports the usual practice in highland *Polylepis* forest restoration projects of only watering saplings at planting and not thereafter. This is very good news as watering in mountainous terrains is hard work, although we point out that watering is probably important for sapling performance in drier environments or in other contexts such as extreme dry periods. Second, saplings benefit from a wind protection shelter. Plastic protections built using discarded water bottles, as the ones used in our study, are cheap to implement and could be important for the early growth of saplings (first few years) exposed to winds. Further measures that could optimize restoration efforts include assessing wind intensity and prevalent wind direction before choosing planting sites in upper topographies and planting at sites more protected from high speed winds whenever possible. Saplings could also possibly benefit from planting at closer distances in areas with intense winds, thus mutually helping to reduce wind effects while growing. The third implication of our study is that all efforts to control and reinforce livestock exclusion are essential to guarantee restoration success in our study area.

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Appendix 1

See Figs. 5 and 6.

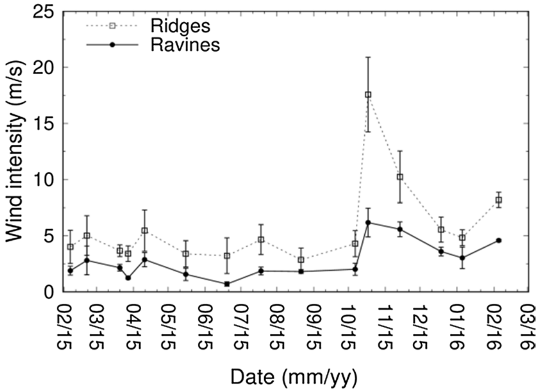

Fig. 5 Average wind intensity for one complete annual cycle. Error bars indicate standard errors

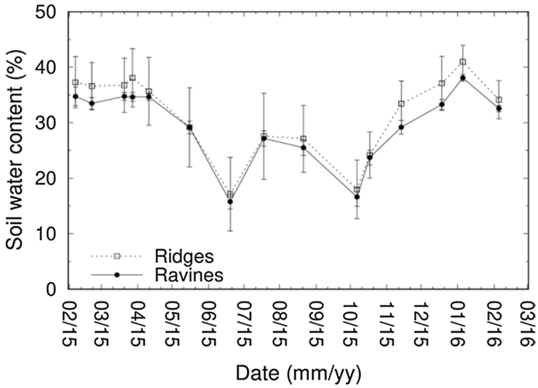

Fig. 6 Average soil water content for one complete annual cycle. Error bars indicate standard errors

Appendix 2

See Table 3.

Table 3 Number of saplings that were alive in March 2017 that had never been recorded as browsed and that were used for re-analyzing the data set without the possible confounding influence of browsing

	Ridges	Ravines
Control with no water addition or wind protection	7	19
Water addition	11	25
Wind protection	25	32
Both water addition and wind protection	31	26

Appendix 3

See Table 4.

Table 4 Results of generalized linear mixed models for growth (assuming a normal distribution) of *P. australis* saplings, testing the effect of fixed factors: topography, water addition treatment (WaT) and wind protection treatment (WiT), as well as all possible double and triple interactions, and including microsite conditions (sun exposure, slope inclination, soil depth, vegetation cover and vegetation height) as covariates

	Growth (cm)	
	t	*P*
Topography	−0.0314	0.9750
WaT	0.8783	0.3805
WiT	3.7908	**0.0002**
Topography × WaT	−0.7157	0.4747
Topography × WiT	−1.2193	0.2237
WaT × WiT	−0.2010	0.8408
Topography × WaT × WiT	0.4964	0.6200
Sun exposure	−4.0438	**0.0001**
Slope Inclination	−0.2780	0.7812
Soil depth	−1.1793	0.2392
Vegetation cover	1.8535	0.0648
Vegetation height	5.0213	**< 0.0001**

Significant results ($P < 0.05$) are shown in bold

References

Alinari J, Von Müller AR, Renison D (2015) The contribution of fire damage to restricting high mountain *Polylepis australis* forests to ravines: insights from an un-replicated comparison. Ecol Austral 25:11–18

Argañaraz JP, Pizarro GG, Zak M, Bellis LM (2015) Fire regime, climate, and vegetation in the Sierras de Córdoba. Fire Ecol 11:55–73

Argibay DS, Renison D (2018) Efecto del fuego y la ganadería en bosques de *Polylepis australis* (Rosaceae) a lo largo de un gradiente altitudinal en las montañas del centro de la Argentina. Bosque 39:145–150

Bergez JE, Dupraz C (2000) Effect of ventilation on growth of *Prunus avium* seedlings grown in treeshelters. Agr For Meteorol 104:199–214

Bond WJ (2005) Large parts of the world are brown or black: a different view on the 'Green World' hypothesis. J Veg Sci 16:261–266

Brooker RW, Maestre FT, Callaway RM, Lortie CL, Cavieres LA, Kunstler G, Liancourt P, Tielbörger K, Travis JM, Anthelme F, Armas C, Coll L, Corcket E, Delzon S, Forey E, Kikvidze Z, Olofsson J, Pugnaire F, Quiroz CL, Saccone P, Schiffers K, Seifan M, Touzard B, Michalet R (2008) Facilitation in plant communities: the past, the present, and the future. J Ecol 96:18–34

Cabido M, Zeballos SR, Zak M, Carranza ML, Giorgis MA, Cantero JJ, Acosta ATR (2018) Native woody vegetation in central Argentina: classification of Chaco and Espinal forests. Appl Veg Sci 21:298–311
Casillo J, Kunst C, Semmartin M (2012) Effects of fire and water availability on the emergence and recruitment of grasses, forbs and woody species in a semiarid Chaco savanna. Austral Ecol 37:452–459
Cingolani AM, Renison D, Zak MR, Cabido MR (2004) Mapping vegetation in a heterogeneous mountain rangeland using landsat data: an alternative method to define and classify land-cover units. Remote Sens Environ 92:84–97
Cingolani AM, Renison D, Tecco P, Gurvich DE, Cabido M (2008) Predicting cover types in a mountain range with long evolutionary grazing history: a GIS approach. J Biogeogr 35:538–551
Cingolani AM, Poca M, Giorgis MA, Vaieretti MV, Gurvich DE, Whitworth-Hulse JI, Renison D (2015) Water provisioning services in a seasonally dry subtropical mountain: identifying priority landscapes for conservation. J Hydrol 525:178–187
Colladon L (2018) Anuario pluviométrico 2012/13–2016/2017, cuenca del Río San Antonio, informe interno del CIRSA. Instituto Nacional del Agua
Colladon L, Pazos I (2014) Anuario pluviométrico 1992/93–2011/12, cuenca del Río San Antonio, 1st edn. Instituto Nacional del Agua, Ezeiza
Colladon L, Pazos I (2016) Anuario de temperatura media mensual 1994–2014, cuenca del Río San Antonio, informe interno del CIRSA. Instituto Nacional del Agua
Cordero RA (1999) Ecophysiology of *Cecropia schreberiana* saplings in two wind regimes in an elfin cloud forest: growth, gas exchange, architecture and stem biomechanics. Tree Physiol 19(3):153–163
Ennos AR (1997) Wind as an ecological factor. Trends Ecol Evol 12(3):108–111
Gardiner B, Berry P, Moulia B (2016) Review: wind impacts on plant growth, mechanics and damage. Plant Sci 245:94–118
Giorgis MA, Cingolani AM, Teich I, Renison D, Hensen I (2010) Do *Polylepis australis* trees tolerate herbivory? Seasonal patterns of shoot growth and its consumption by livestock. Plant Ecol 207:307–319
Grace J (1988) Plant response to wind. Agric Ecosyst Environ 22(23):71–88
Hammatt N (1998) Influence of tree shelters, irrigation and branch pruning on early field performance of micropropagated wild cherry cv. F12/1 New For 15:261–269
Holtmeier F, Broll G (2010) Wind as an ecological agent at treelines in North America, the Alps, and the European Subarctic. Phys Geog 31(3):203–233
Kessler M (2002) The “*Polylepis* problem”: where do we stand? Ecotropica 8:97–110
Kjelgren R, Rupp LA (1997) Establishment in treeshelters I: shelters reduce growth, water use, and hardiness, but not drought avoidance. HortScience 32(7):1281–1283
Körner C (2012) Alpine treelines: functional ecology of the global high elevation tree limits, 1st edn. Springer, Basel
Kronfuss H, Havranek WM (1999) Effects of elevation and wind on the growth of *Pinus cembra* L. in a subalpine afforestation. Phyton (Austria; special issue “Eurosilva”) 39:99–106
Lawton RO (1982) Wind stress and elfin stature in a montane rain forest tree: an adaptive explanation. Am J Bot 69:1224–1230
Littell JS, Peterson DL, Tjoelker M (2008) Douglas-fir growth in mountain ecosystems: water limits tree growth from stand to region. Ecol Monogr 78:349–368
Loranger H, Zotz G, Bader MY (2017) Competitor or facilitator? The ambiguous role of alpine grassland for the early establishment of tree seedlings at treeline. Oikos 126:1625–1636
McArthur C, Bradshaw OS, Jordan GJ, Clissold FJ, Pile AJ (2010) Wind affects morphology, function, and chemistry of eucalypt tree seedlings. Int J Plant Sci 171:73–80
Oliet JA, Jacobs DF (2007) Microclimatic conditions and plant morpho-physiological development within a tree shelter environment during establishment of *Quercus ilex* seedlings. Agr For Meteorol 144:58–72
Orrego F (2011) Queñoa: Árbol de las alturas. Gerencia de Medio Ambiente de la Compañía Minera Doña Inés de Collahuasi, Santiago
Padilla FM, Pugnaire FI (2006) The role of nurse plants in the restoration of degraded environments. Front Ecol Environ 4:196–202
Pia MV, Renison D, Mangeaud A, De Angelo C, Haro JG (2013) Occurrence of top carnivores in relation to land protection status, human settlements and rock outcrops in the high mountains of central Argentina. J Arid Environ 91:31–37
Pollice J, Marcora P, Renison D (2013) Seed production in *Polylepis australis* (Rosaceae) as influenced by tree size, livestock and interannual climate variations in the mountains of central Argentina. New For 44:233–247
Potter MJ (1991) Tree shelters. Forestry Commission. Handbook 7, HM Stationery Office, London
R Core Team (2018) R: a language and environment for statistical computing. R Foundation for Statistical Computing, Vienna. https://www.R-project.org/

Rada F, García-Núñez C, Rangel S (2009) Low temperature resistance in saplings and ramets of *Polylepis sericea* in the Venezuelan Andes. Acta Oecol 35:610–613

Renison D, Cingolani AM, Schinner D (2002) Optimising restoration of *Polylepis australis* forests through reforestation: when, where and how to transplant seedlings to the mountains? Ecotropica 8:219–224

Renison D, Cingolani AM, Suarez R, Menoyo E, Coutsiers C, Sobral A, Hensen I (2005) The restoration of degraded mountain woodlands: effects of seed provenance and microsite characteristics on *Polylepis australis* seedling survival and growth in central Argentina. Restor Ecol 13:129–137

Renison D, Hensen I, Suarez R, Cingolani AM (2006) Cover and growth habit of *Polylepis* woodlands and shrublands in the mountains of Central Argentina: human or environmental influence? J Biogeogr 33:876–887

Renison D, Hensen I, Suarez R (2011) Landscape structural complexity of high-mountain *Polylepis australis* forests: a new aspect of restoration goals. Restor Ecol 19:390–398

Renison D, Chartier MP, Menghi M, Marcora PI, Torres RC, Giorgis M, Hensen I, Cingolani AM (2015) Spatial variation in tree demography associated to domestic herbivores and topography: insights from a seeding and planting experiment. For Ecol Manag 335:139–146

Renison D, Morales L, Cuyckens E, Sevillano CS, Cabrera-Amaya DM (2018) Ecología y conservación de los bosques y arbustales de *Polylepis*: ¿Qué sabemos y qué ignoramos? Ecol Austral 28:163–174

Ripple WJ, Beschta RL (2003) Wolf reintroduction, predation risk, and cottonwood recovery in Yellowstone National Park. For Ecol Manag 184:299–313

Schönbeck L, Lohbeck M, Bongers F, Ramos M, Sterck F (2015) How do light and water acquisition strategies affect species selection during secondary succession in moist tropical forests? Forests 6:2047–2065

Segovia-Salcedo MC, Domic A, Boza T, Kessler M (2018) Situación taxonómica de las especies del género Polylepis. Implicancias para los estudios ecológicos, la conservación y la restauración de sus bosques. Ecol Austral 28:188–201

Simoes Macayo N, Renison D (2015) ¿Cuántos años monitorear el éxito de plantaciones con fines de restauración? Análisis en relación al micrositio y procedencia de las semillas. Bosque 36:315–322

Sørensen R, Zinko U, Seibert J (2006) On the calculation of the topographic wetness index: evaluation of different methods based on field observations. Hydrol Earth Syss Sci 10:101–112

Suarez ML, Renison D, Hensen I, Marcora P (2008) Age-size-habitat relationships for *Polylepis australis*: dealing with endangered forest ecosystems. Biodivers Conserv 17:2617–2625

Von Müller AR, Cingolani AM, Vaieretti MV, Renison D (2012) Estimation of localized cattle stocking rate from dung frequency in a mountain grassland. Ecol Austral 22:178–187

Whittaker RH (1975) Communities and ecosystems, 2nd edn. Collier MacMillan, London